

Дз с 26.09 на 03.10

Множества и отношения

Задача 1.1

$$A = \{1, 2, 3, 4, 5, 6, 7\} \quad B = \{4, 5, 6, 7, 8, 9, 10\} \quad C = \{2, 4, 6, 8, 10\}$$

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

a) $A \cup C = \{1, 2, 3, 4, 5, 6, 7, 8, 10\}$

б) $A \cap B = \{4, 5, 6, 7\}$

в) $A \cap (B \cup C) = ?$

□ $B \cup C = \{2, 4, 5, 6, 7, 8, 9, 10\}$

$$A \cap (B \cup C) = \{2, 4, 5, 6, 7\}$$

г) $(A \cap B) \cup C = ?$

□ $A \cap B = \{4, 5, 6, 7\}$

$$(A \cap B) \cup C = \{2, 4, 5, 6, 7, 8, 10\}$$

д) $(A \cap B)' = ?$

□ $A \cap B = \{4, 5, 6, 7\}$

$$(A \cap B)' = U \setminus (A \cap B) = \{1, 2, 3, 8, 9, 10\}$$

$$e) A' \cap B' = ?$$

$$\square A' = U \setminus A = \{8, 9, 10\}$$

$$B' = U \setminus B = \{1, 2, 3\}$$

$$A' \cap B' = \emptyset$$

$$2re) A \Delta B = \{1, 2, 3, 8, 9, 10\}$$

$$3) A - B = \{1, 2, 3\}$$

Задача 1.2

$$A = \{1, 2, 3, 4, 5, 6, 7\} \quad B = \{4, 5, 6, 7, 8, 9, 10\}$$

$$C = \{2, 4, 6, 8, 10\} \quad U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$a) A - C = \{1, 3, 5, 7\}$$

$$б) (A - B) \cup (B - A) = ?$$

$$\square A - B = \{1, 2, 3\}$$

$$B - A = \{8, 9, 10\}$$

$$(A - B) \cup (B - A) = \{1, 2, 3, 8, 9, 10\}$$

$$в) A \cap (B \cap C') = ?$$

$$\square C' = U \setminus C = \{1, 3, 5, 7, 9\}$$

$$B \cap C' = \{5, 7, 9\}$$

$$A \cap (B \cap C') = \{5, 7\}$$

$$z) (A \cup C) - B' = ?$$

$$\square A \cup C = \{1, 2, 3, 4, 5, 6, 7, 8, 10\}$$

$$B' = U \setminus B = \{1, 2, 3\}$$

$$(A \cup C) - B' = \{4, 5, 6, 7, 8, 10\}$$



$$g) (A - \emptyset) \cup (A - A)$$

$$\square A - \emptyset = \{1, 2, 3, 4, 5, 6, 7\}$$

$$A - A = \emptyset$$

$$(A - \emptyset) \cup (A - A) = \{1, 2, 3, 4, 5, 6, 7\}$$



$$e) B \Delta C = \{2, 5, 7, 9\}$$

$$xc) C - A = \{8, 10\}$$